Cadence Tempus STA Physical-Fix Labs
Beginner Self-Study & Solution Manual

Independent ready-to-run labs • Netlist • SDC • Tcl • FAIL/FIX calculations

1. How to use this package

Each lab is completely self-contained. Start in the selected LABxx directory. First inspect the Liberty file, then the FAIL netlist, FAIL SDC and FAIL Tcl script. Predict the result by hand. Run Tempus. Next compare the fixed netlist/SDC, predict the new result, and run the fixed script. Do not judge a timing fix only by whether the report is green; understand whether the netlist, constraint, or timing exception changed and why.

2. Folder structure used in every lab

lib/ contains the local Liberty timing library. netlist/ contains 01_fail.v and 02_fixed.v. sdc/ contains the constraints. scripts/ contains the Tempus Tcl flow. reports/ receives logs from run_all.sh. No lab sources a file from another lab.

Run individually: tempus -files scripts/01_run_fail.tcl and tempus -files scripts/02_run_fixed.tcl
Run both: ./run_all.sh

3. Timing values in the teaching Liberty

Cell / arc	Teaching delay	Purpose
BUF_X1 A→Y	0.50 ns	Slow buffer
BUF_X2 A→Y	0.25 ns	Faster/up-sized buffer
DLY_X1 A→Y	1.00 ns	Intentional delay cell
DFFPOS/DFFNEG CLK→Q	0.20 ns	Clock-to-Q
DFF setup	0.20 ns	Setup requirement
DFF hold	0.10 ns	Intrinsic hold requirement

Important — how Tempus actually computes the checks: a register-to-register path is always timed from the launch CLOCK edge, so the launch flip-flop CLK→Q delay (0.20 ns) is part of the data arrival time. Restricting report_timing to start at a Q pin does NOT remove it; the report's first data line is the CLK→Q arc. The scripts therefore use -from <launch>/CLK and every hand calculation below includes CLK→Q. With ideal clocks (no latency, no parasitics) Tempus uses:

SETUP (report_timing -late): Arrival = CLK→Q + Σ cell delays. Required = Phase shift − Setup(lib) − setup uncertainty, where Phase shift = capture edge − launch edge. Slack = Required − Arrival.

HOLD (report_timing -early): Arrival = CLK→Q + Σ cell delays. Required = Hold phase shift + Hold(lib) + hold uncertainty, where the hold capture edge is one period before the setup capture edge (0 ns for same-edge paths). Slack = Arrival − Required. With set_min_delay the min-delay value replaces the hold phase shift and the library hold is still added.

These are exactly the Tempus report fields: Other End Arrival Time, − Setup / + Hold, + Phase Shift, − Uncertainty, = Required Time, Arrival Time, Slack Time.

4. Tcl versus SDC

Tcl controls the Tempus flow: read_lib, read_verilog, set_top_module, read_sdc, report_timing and exit. SDC describes timing intent: create_clock, set_clock_uncertainty, set_min_delay and set_false_path. In this package the SDC commands are stored in separate .sdc files so beginners can distinguish design constraints from tool-flow commands. Timing exceptions use a valid startpoint (-from [get_cells U_LAUNCH], i.e. the launch FF clock pin), not a Q pin.

LAB01 — Setup Violation and Physical Fix

Netlist concept

Positive-edge launch FF → eight BUF_X1 cells → positive-edge capture FF. The fixed netlist preserves topology but changes all eight buffers to BUF_X2.

Constraint and timing requirement

Clock = 4.0 ns (ideal). Setup required time = 4.00 − 0.20 = 3.80 ns. Hold required time = 0.00 + 0.10 = 0.10 ns.

Why the original case fails

FAIL: Arrival = CLK→Q + 8 × 0.50 = 0.20 + 4.00 = 4.20 ns. Setup slack = Required − Arrival = 3.80 − 4.20 = −0.40 ns (VIOLATED). Hold slack = Arrival − Required = 4.20 − 0.10 = +4.10 ns (MET). Data is late, so setup fails.

How the correction works

FIX: replace BUF_X1 by BUF_X2. Arrival = 0.20 + 8 × 0.25 = 2.20 ns. Setup slack = 3.80 − 2.20 = +1.60 ns (MET); improvement = 2.00 ns. Hold slack = 2.20 − 0.10 = +2.10 ns (still MET, but lower — speeding up a path always eats hold margin). The clock constraint is unchanged.

Rule: setup failure means the data path is too slow; reduce maximum data-path delay.

LAB02 — Minimum-Delay / Hold-Style Physical Fix

Netlist concept

Positive-edge launch FF → two BUF_X1 cells → positive-edge capture FF. The fixed netlist inserts two DLY_X1 cells after the two buffers.

Constraint and timing requirement

The SDC sets set_min_delay 2.50 -from [get_cells U_LAUNCH] -to U_CAPTURE/D. For a flip-flop endpoint Tempus replaces the normal 0 ns hold relationship with 2.50 ns and still adds the library hold time: hold required = 2.50 + 0.10 = 2.60 ns. Do not confuse the 2.50 ns teaching requirement with the 0.10 ns library hold (without the min delay the plain hold check would pass). The setup check still uses the 10 ns period: setup required = 10.00 − 0.20 = 9.80 ns.

Why the original case fails

FAIL: Arrival = CLK→Q + 2 × 0.50 = 0.20 + 1.00 = 1.20 ns. Hold (min-delay) slack = Arrival − Required = 1.20 − 2.60 = −1.40 ns (VIOLATED). Setup slack = 9.80 − 1.20 = +8.60 ns (MET). The path is too fast; at least 1.40 ns of extra minimum delay is needed.

How the correction works

One DLY_X1 would give arrival 2.20 ns and still fail by 2.20 − 2.60 = −0.40 ns. Two DLY_X1 cells give arrival 0.20 + 1.00 + 2 × 1.00 = 3.20 ns. Hold slack = 3.20 − 2.60 = +0.60 ns (MET). Setup slack = 9.80 − 3.20 = +6.60 ns (still MET).

Rule: a minimum-delay/hold problem is repaired by increasing minimum data-path delay; do not weaken a valid minimum-delay constraint.

LAB03 — Setup with Clock Uncertainty / Margin

Netlist concept

Positive-edge launch FF → eight BUF_X1 cells → positive-edge capture FF. The fixed netlist replaces all eight with BUF_X2.

Constraint and timing requirement

Clock = 4.4 ns, set_clock_uncertainty -setup 0.4 ns, setup = 0.20 ns. Setup required time = 4.40 − 0.40 − 0.20 = 3.80 ns. The uncertainty is -setup only, so hold required = 0.00 + 0.10 + 0.00 = 0.10 ns.

Why the original case fails

FAIL: Arrival = 0.20 + 8 × 0.50 = 4.20 ns. Setup slack = 3.80 − 4.20 = −0.40 ns (VIOLATED). Hold slack = 4.20 − 0.10 = +4.10 ns (MET).

How the correction works

FIX: keep the 0.4 ns uncertainty and use BUF_X2. Arrival = 0.20 + 8 × 0.25 = 2.20 ns. Setup slack = 3.80 − 2.20 = +1.60 ns (MET). Hold slack = 2.20 − 0.10 = +2.10 ns (MET). Removing uncertainty would be an artificial constraint relaxation, not a physical fix.

Rule: preserve valid clock margin and optimize the data path.

LAB04 — Half-Cycle Setup Physical Fix

Netlist concept

DFFNEG_X1 launch FF → eight buffers → DFFPOS_X1 capture FF. The clock is 8 ns with waveform {0 4}. Launch is on the 4 ns falling edge and capture is on the next 8 ns rising edge.

Constraint and timing requirement

Setup: launch on the falling edge at 4 ns, capture on the next rising edge at 8 ns, so Tempus phase shift = 8 − 4 = 4.00 ns, not 8 ns. Setup required = 4.00 − 0.20 = 3.80 ns (relative to the launch edge). Hold: the hold capture edge is the rising edge at 0 ns (one period before 8 ns), so hold phase shift = 0 − 4 = −4.00 ns and hold required = −4.00 + 0.10 = −3.90 ns.

Why the original case fails

FAIL: Arrival = 0.20 + 8 × 0.50 = 4.20 ns after the launch edge (absolute 8.20 ns vs. required 7.80 ns). Setup slack = 3.80 − 4.20 = −0.40 ns (VIOLATED). Hold slack = 4.20 − (−3.90) = +8.10 ns (MET).

How the correction works

FIX: replace BUF_X1 with BUF_X2. Arrival = 0.20 + 8 × 0.25 = 2.20 ns. Setup slack = 3.80 − 2.20 = +1.60 ns (MET). Hold slack = 2.20 + 3.90 = +6.10 ns (MET). Clock period and waveform remain unchanged.

Rule: for opposite-edge paths, identify the actual launch and capture edges first; never automatically use the full clock period.

LAB05 — Correct False-Path Timing Exception

Netlist concept

The netlist has two branches. DATA: U_DATA_L → six BUF_X1 → U_DATA_C. CFG: U_CFG_L → twelve BUF_X1 → U_CFG_C. The same netlist is used before and after the exception.

Constraint and timing requirement

Clock = 5.0 ns: setup required = 5.00 − 0.20 = 4.80 ns, hold required = 0.10 ns. DATA arrival = 0.20 + 6 × 0.50 = 3.20 ns → setup slack +1.60 ns, hold slack +3.10 ns. CFG arrival = 0.20 + 12 × 0.50 = 6.20 ns → setup slack 4.80 − 6.20 = −1.40 ns if timed, hold slack +6.10 ns.

Why the original case fails

BEFORE: the CFG path is timed and appears as a −1.40 ns setup violation; the DATA path passes.

How the correction works

AFTER: set_false_path -from [get_cells U_CFG_L] -to [get_pins U_CFG_C/D] excludes the functionally false CFG path from both setup and hold analysis. Its arrival is still 6.20 ns; it does not acquire positive slack. The normal DATA path remains timed (+1.60 ns setup, +3.10 ns hold).

Rule: never declare a path false merely because it fails. Apply set_false_path only when functional design intent proves the normal timing relationship does not apply.

10. Consolidated expected hand calculations

Lab	Case	Arrival (CLK→Q + data)	Setup: required / slack	Hold: required / slack	Action
LAB01	FAIL	0.20+4.00 = 4.20	3.80 / −0.40 FAIL	0.10 / +4.10	Speed up
LAB01	FIX	0.20+2.00 = 2.20	3.80 / +1.60	0.10 / +2.10	BUF_X2
LAB02	FAIL	0.20+1.00 = 1.20	9.80 / +8.60	2.60 / −1.40 FAIL	Add delay
LAB02	FIX	0.20+3.00 = 3.20	9.80 / +6.60	2.60 / +0.60	2×DLY_X1
LAB03	FAIL	0.20+4.00 = 4.20	3.80 / −0.40 FAIL	0.10 / +4.10	Keep uncertainty
LAB03	FIX	0.20+2.00 = 2.20	3.80 / +1.60	0.10 / +2.10	BUF_X2
LAB04	FAIL	0.20+4.00 = 4.20	3.80 / −0.40 FAIL	−3.90 / +8.10	Half-cycle
LAB04	FIX	0.20+2.00 = 2.20	3.80 / +1.60	−3.90 / +6.10	BUF_X2

LAB05	BEFORE CFG	0.20+6.00 = 6.20	4.80 / −1.40 FAIL	0.10 / +6.10	Check intent
LAB05	AFTER CFG	0.20+6.00 = 6.20	False path	False path	SDC exception

11. How to read a Tempus timing report

First identify the startpoint and endpoint. Next identify whether the report is late/max (setup) or early/min (hold/minimum delay). Read the arrival time, required time and slack. Negative slack means the analyzed requirement is violated. Then determine whether the correct remedy is a physical netlist change, a legitimate clock/data constraint, or a justified timing exception. Never modify constraints only to make negative slack disappear.

12. Beginner troubleshooting checklist

- Run Tempus from the LABxx directory so relative paths resolve consistently.
- Confirm lib/teaching_sta_physical.lib exists locally in that lab.
- Confirm the top module in the Tcl matches the Verilog module name.
- Confirm SDC object names such as U_LAUNCH and U_CAPTURE/D exactly match the netlist instances/pins.
- If a report differs from the hand arithmetic, compare it field by field: Phase Shift, Setup/Hold, Uncertainty, Required Time, and the CLK→Q line at the top of the data path.
- LAB05 is an exception demonstration: after set_false_path, do not expect the excluded CFG path to show a new positive slack.

13. Recommended self-study sequence

For every lab: (1) draw the FF-to-FF path, (2) list the cell delays from the Liberty, (3) write the SDC equation, (4) calculate expected FAIL slack, (5) inspect the fixed netlist or exception, (6) calculate the expected fixed result, (7) run Tempus, and (8) explain in one sentence why the correction is physically or functionally valid.

14. Quick viva questions

Q: Why does LAB01 fail?
A: The maximum data path is 4.20 ns (0.20 CLK→Q + 4.00) but the setup required time is 3.80 ns, giving −0.40 ns slack.

Q: Why add delay in LAB02?
A: A minimum-delay/hold-style failure means data arrives too early; increasing minimum path delay fixes it.

Q: Why not delete uncertainty in LAB03?
A: That relaxes a valid margin instead of improving the circuit.

Q: Why does LAB04 use only 4 ns of an 8 ns clock?
A: Launch is at the 4 ns falling edge and capture at the next 8 ns rising edge.

Q: Does set_false_path speed up LAB05?
A: No. The physical path remains 6.20 ns (CLK→Q + 6.00); the justified path is excluded from STA.

15. Terminal calculation output built into every Tcl script

Every FAIL and FIX Tcl script prints an EXPECTED HAND CALCULATION block before invoking report_timing. For both the SETUP and HOLD check it shows CLK→Q, the cell delays, launch and capture edges, phase shift, setup/hold and uncertainty terms, required time, arrival time and slack — in the same order as the Tempus report fields. It then runs report_timing -late (setup) and report_timing -early (hold) with -path_type full_clock so the clock edges are visible.

For LAB05, the fixed script does not claim that the false CFG path receives positive slack. It prints that the CFG arrival remains 6.20 ns and would still have −1.40 ns setup slack if timed normally; the justified set_false_path removes that path from setup and hold analysis while the normal DATA path remains timed.

Recommended student practice: read the printed equation first, predict the report, then compare the Tempus arrival/required/slack fields. If the installed Tempus release reports a slightly different value, inspect exactly which clock, setup, launch/capture and path terms are included before changing any constraint.